

Year 10 Pathways 1

Mini Test 1

Name: MARKER GUIDE

Total Marks: 30 marks
Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1

3 marks

Match the definition on the left-hand side with the correct answer on the right-hand side by drawing a line between each one:

non-recurring decimal	highest common factor
LCM	An irrational number that is the root of a rational number, particularly an integer.
mixed number	a real number that can be written as a ratio of two integers or a fraction excluding zero as a denominator
Rational number	A decimal that does not recur.
HCF	least common multiple
surd	number written as a whole number and a fraction

Question 2

1 mark

Express each of the following in standard forms.

(a) 629000

$$6.29 \times 10^5$$

(b) 0.0000072453

$$7.2453 \times 10^{-6}$$

Question 3

3 marks

Evaluate, giving answers in simplest form without indices.

(i) $(3^2)^3$

$$= 3^6$$

$$= \underline{729}$$

(ii) $5^{-1} + 5^2$

$$= \frac{1}{5} + 25$$

$$= \underline{25\frac{1}{5}}$$

(iii) $\frac{1}{9} \times (3^{-2})^{-4}$

$$= \frac{3^8}{3^2}$$

$$= 3^6$$

$$= \underline{729}$$

Question 4

8 marks

Simplify each of the following expressing answers with positive indices where appropriate.

(a) $\frac{16c^3e^4}{24ce^3}$

$$= \frac{2c^2e}{3}$$

(b) $\frac{a^{-3}b^2}{b^5}$

$$= \frac{1}{a^3b^3}$$

(c) $(2xy^{-3})^2$

$$\frac{4x^2}{y^6}$$

(d) $(p^2)^0$

$$= \underline{1}$$

(e) $4x^7 \cdot 3c^{-2}$

$$= \frac{12x^7}{c^2}$$

Question 5

2 marks

Solve each of the following equations.

(a) $3^x = 27$

$$3^x = 3^3$$

$$\therefore \underline{x = 3}$$

(b) $4^{3x} = 8^{x-2}$

$$(2^2)^{3x} = 2^{3(x-2)}$$

$$\therefore 6x = 3x - 6$$

$$3x = -6$$

$$\underline{x = -2}$$

Question 6
1 mark

Circle the integers in the following list.

[1 mark]

$\sqrt[3]{80}$

$\sqrt{49}$

$\sqrt{169}$

$\sqrt{48}$

$\sqrt{10000}$

$\sqrt[3]{8}$

Question 7
4 marks

Simplify each of the following leaving your answer in surd form.

(a) $\sqrt{6} + 3\sqrt{6}$

$= 4\sqrt{6}$

(b) $\sqrt{32} + \sqrt{8}$

$$= 4\sqrt{2} + 2\sqrt{2}$$

$$= 6\sqrt{2}$$

(c) $6\sqrt{5} \cdot 5\sqrt{6}$

$= 30\sqrt{30}$

Question 8
3 marks

Rationalise the following

(a) $\frac{\sqrt{2}}{\sqrt{7}}$

$$= \frac{\sqrt{14}}{7}$$

(b) $\frac{1}{2\sqrt{6}}$

$$= \frac{\sqrt{6}}{12}$$

Question 9
5 marks

(a) Simplify $\frac{\log_5 125}{\log_5 \frac{1}{25}}$

$$= \frac{\log_5 5^3}{\log_5 5^{-2}} = \frac{3}{-2}$$

(b) Calculate $\log_2 14 - \log_2 21 + \log_2 6$

$$= \log_2 \left[\frac{14}{21} \cdot 6 \right]$$

$$= \log_2 \left(\frac{12}{3} \right) = \log_2 2^2 = 2$$

(c) Find exactly $\log_3 81 + \log_2 0.5$

$$= \log_3 3^4 + \log_2 2^{-1}$$

$$= 4 - 1$$

$$= 3$$

(d) What is the value of $\log 1000$?

$$= 3$$

(e) Evaluate $\log \sqrt[4]{100}$

$$= \log_{10} (10^2)^{\frac{1}{4}}$$

$$= \frac{1}{2}$$